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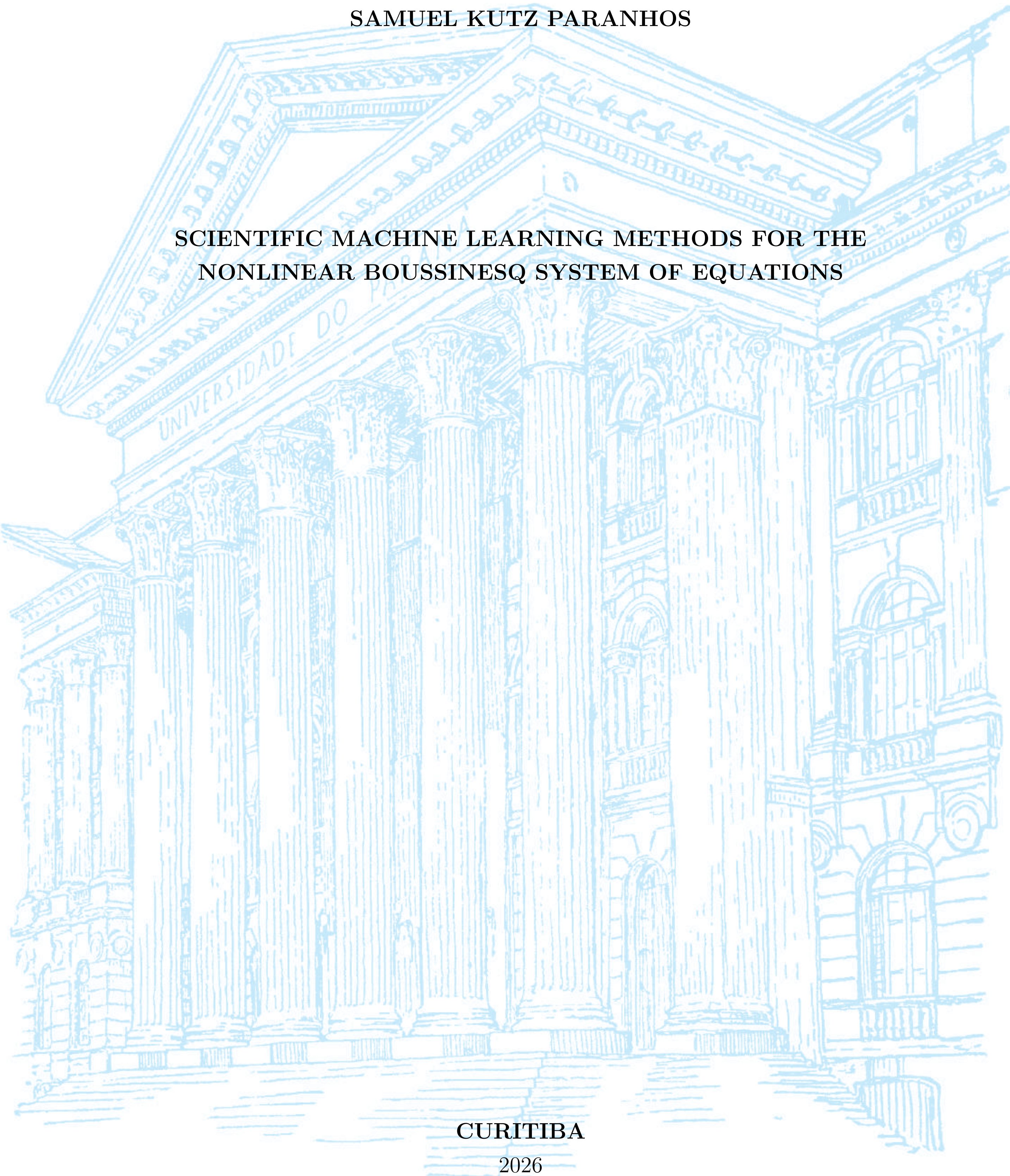

SCIENTIFIC MACHINE LEARNING METHODS FOR THE NONLINEAR BOUSSINESQ SYSTEM OF EQUATIONS

CURITIBA

2026

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TRABALHO DE CONCLUSÃO DE CURSO

SCIENTIFIC MACHINE LEARNING METHODS FOR THE NONLINEAR BOUSSINESQ SYSTEM OF EQUATIONS

Trabalho apresentado como requisito parcial à obtenção do grau de Bacharel em Matemática Industrial no Departamento de Matemática da Universidade Federal do Paraná.

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RESUMO

Este trabalho investiga a eficácia de métodos de Scientific Machine Learning (SciML) na resolução e aproximação do sistema não linear de Boussinesq, um sistema de Equações Diferenciais Parciais (EDPs) utilizado para modelar a dinâmica de ondas aquáticas. Neste estudo, o sistema de Boussinesq será tratado como uma EDP dependente de dois parâmetros: os parâmetros de não linearidade e de dispersão. O objetivo é analisar o uso do ecossistema contemporâneo de ferramentas desenvolvido em torno do recente crescimento de Machine Learning como complemento aos métodos numéricos clássicos empregados na solução do sistema de Boussinesq. Foi utilizado o método pseudoespectral para gerar um conjunto de dados consistindo de soluções de referência do sistema de Boussinesq para diversos parâmetros e é então empregado como base de treinamento para três arquiteturas de SciML investigadas neste trabalho: Redes Neurais Informadas pela Física (PINNs), treinadas a um único parâmetro tanto com restrições de dados quanto de forma puramente física; Operadores Neurais de Fourier (FNO), operando em regime puramente supervisionado a vários parâmetros com uso exclusivo de dados sem nenhuma informação do operador diferencial da EDP; e Operadores Neurais Informados pela Física (PINO), que consiste em uma FNO com a inclusão do operador diferencial da EDP na função de perda da arquitetura. Como métricas de eficiência dessas arquiteturas, utilizamos a evolução do erro espectral e do erro relativo ao longo do espaço-tempo com relação a solução de referência. Como resultados, percebe-se que as PINNs tradicionais, especialmente quando treinadas sem o auxílio de dados, sofrem de um fenômeno conhecido como viés espectral, apresentando tendência a aprender primeiramente as componentes de baixa frequência da solução e grandes dificuldades em capturar detalhes essenciais de alta frequência à medida que a não linearidade aumenta. Em contrapartida, por aprenderem diretamente no espaço das frequências, os modelos de aprendizado de operadores, como o FNO e o PINO treinado com dados, não exibem esse tipo de limitação, a qual reaparece apenas quando o PINO é treinado de forma puramente física. Assim, concluímos que a inclusão de dados na função de perda de redes neurais puramente físicas constitui uma estratégia eficaz para mitigar o viés espectral comumente presente nessas arquiteturas.

Palavras-chave: Scientific Machine Learning; Equações Diferenciais Parciais; Sistema de Boussinesq; Redes Neurais Informadas pela Física.

ABSTRACT

This work investigates the effectiveness of Scientific Machine Learning (SciML) methods in solving and approximating the nonlinear Boussinesq system, a system of Partial Differential Equations (PDEs) used to model water wave dynamics. In this study, the Boussinesq system will be treated as a PDE dependent on two parameters: the nonlinearity and dispersion parameters. The objective is to analyze the use of the contemporary ecosystem of tools developed around the recent growth of Machine Learning as a complement to the classical numerical methods employed in solving the Boussinesq system. The pseudospectral method was used to generate a dataset consisting of reference solutions of the Boussinesq system for various parameters and is then employed as the training basis for three SciML architectures investigated in this work: Physics-Informed Neural Networks (PINNs), trained at a single parameter both with data constraints and purely physics-informed; Fourier Neural Operators (FNOs), operating in a purely supervised regime at multiple parameters using only data without any differential operator information from the PDE; and Physics-Informed Neural Operators (PINOs), which consist of an FNO with the inclusion of the PDE differential operator in the architecture loss function. As efficiency metrics for these architectures, we use the evolution of spectral error and relative error over space-time with respect to the reference solution. As results, it is observed that traditional PINNs, especially when trained without the aid of data, suffer from a phenomenon known as spectral bias, showing a tendency to learn the low-frequency components of the solution first and great difficulty capturing essential high-frequency details as nonlinearity increases. In contrast, by learning directly in the frequency space, operator learning models such as FNO and PINO trained with data do not exhibit this type of limitation, which reappears only when PINO is trained in a purely physics-informed manner. Thus, we conclude that including data in the loss function of purely physics-informed neural networks constitutes an effective strategy to mitigate the spectral bias commonly present in these architectures.

Keywords: Scientific Machine Learning; Partial Differential Equations; Boussinesq System; Physics-Informed Neural Networks.

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1 INTRODUCTION

Since the start of the scientific thinking, mathematics has been one of the key languages in which we can understand the world, and Partial Differential Equations (PDEs) are one of the main dialects of the mathematical language we use to model and to simplify nature's complexity.

When modelling these equations, we cannot always afford them to have simple closed-form solutions (that is, when they have solution at all), this exposes the need of obtaining approximated solutions coming from numerical methods for these PDEs, also giving the option to circumvent analytical complexity. This allows the extraction of predictions, comparisons with data, build simulations and so on. These classical numerical methods, widely studied for all types of differential equations, usually rely on discretizing the PDE's domain into grids and performing temporal steps.

With the recent ascension of Machine Learning (ML) methods and a whole ecosystem of tools built around it, a natural path is to bridge these two worlds and let ML augment the numerical solution of PDEs, taking advantage of the many software frameworks developed. This is one facet of Scientific Machine Learning (SciML), the broader effort to fuse physics-based, mechanistic models with data-driven methods. Within SciML, this work focuses on the numerical solution of PDEs and, given the vast literature around the topic, restricts its review of recent works to hydrodynamics equations.

1.1 Literature Overview on Scientific Machine Learning for PDEs

The neural network building blocks underlying all of these approaches trace a line beginning with the formal neuron of McCulloch et al. (1943) and Rosenblatt's perceptron (ROSENBLATT, 1958), the first trainable single-layer model. The key algorithmic advance that unlocked multi-layer training was the backpropagation algorithm (RUMELHART et al., 1986), making gradient-based optimization practical for deep architectures. Shortly after, the universal approximation theorem (CYBENKO, 1989; HORNIK et al., 1989) provided the theoretical guarantee for a multilayer perceptron (MLP) with a single hidden layer of sufficient width to be able to approximate any continuous function on a compact domain to arbitrary precision. This combination of practical and universal approximation established the MLP as the backbone of modern deep learning. The term neural network itself reflects the biological metaphor at the heart of these models, in which McCulloch et al. (1943) drew an explicit analogy to the firing of biological neurons, and this vocabulary propagated through the field, causing MLPs to be routinely and interchangeably referred to as artificial neural networks (ANNs) or feedforward neural networks. The idea of “deep” in deep learning, popularized following the breakthrough results of LeCun et al. (2015), simply denotes an MLP with many hidden layers, a regime in which representational capacity grows substantially and where the backpropagation-based training paradigm proves to be exceptionally efficient.

These ML foundations become the basis of a scientific methodology once the learner is coupled to the physical laws that govern a problem of interest. This is the premise of Scientific Machine Learning, a term consolidated by Baker et al. (2019) to name the discipline at the intersection of scientific computing and machine learning, and surveyed in breadth by Karniadakis et al. (2021). Its unifying goal is to combine the interpretability, generalization, and inductive biases of physics-based models with the flexibility and data-adaptivity of ML, rather than treating the network as a pure black box. The methods span several families. Some recover governing equations directly from data, as in the Sparse Identification of Nonlinear Dynamics (SINDy) (BRUNTON et al., 2016). Others embed differential-equation structure into the network itself, as in Neural Ordinary Differential Equations (neural ODEs) (CHEN, R. T. Q. et al., 2018) and Universal Differential Equations (UDEs), in which a trainable network stands in for the unknown terms of an otherwise mechanistic model (RACKAUCKAS et al., 2020). Structure-preserving, or physics-encoded, architectures instead hard-wire physical invariants, as with Hamiltonian Neural Networks that conserve energy by construction (GREYDANUS et al., 2019), while graph-based simulators learn the dynamics of discretized systems directly from mesh or particle data (SANCHEZ-GONZALEZ et al., 2020). These tools have been carried across a wide range of domains, from global weather and climate forecasting (BONEV et al., 2023) to molecular dynamics, materials design, and turbulence modelling (KARNIADAKIS et al., 2021). Among all these directions, the one that concerns us here is the use of SciML to obtain numerical solutions of PDEs and, more specifically, of the hydrodynamics equations that govern water waves; the remainder of this review is devoted to that subarea.

The contemporary landscape of SciML for PDEs was reshaped by one idea in particular, the Physics-Informed Neural Networks (PINNs) (RAISSI et al., 2019), that is, the embedding of the governing PDE residual directly into the training loss of a Neural Network, working as regularization term for not only fitting data, but also respecting the underlying physics. In hydrodynamics, there is a growing body of work has applied

PINNs to: water flow (DE PINA et al., 2021, 2023), shallow-water dynamics on the sphere (BIHLO et al., 2022), and inverse problems in strongly nonlinear wave systems (JAGTAP et al., 2022). The consistent result that we can address from reviews (CUOMO et al., 2023) is that PINNs are most effective when observational data are sparse and the task is parameter identification, namely inverse problems, which are known for usually being not well-posed probelms. But, for a reliable precise forward simulation at scale, they do not yet rival classical solvers. In our vision, these approaches serve best to complement them.

As a natural extension from neural networks, there is a recent development in learning maps between infinite-dimensional function spaces, in which the theoretical foundation was popularized by T. Chen et al. (1995), who extended the universal approximation theorem to nonlinear operator, in other words, a shallow network can approximate any continuous operator between Banach spaces to arbitrary precision. Bringing this idea into practical terms, Lu et al. (2021) introduced DeepONet, the first deep architecture expressly designed for learning operators coming from differential equations, demonstrating it across a range of ODEs and PDEs. Building on this line of work, operator learning puts aside the single-instance limitation of PINNs by training a model to approximate the full mapping from parameters and initial data to the solution, so that at inference time a new solution costs only a forward pass for the network. Parallel to DeepONet, Fourier Neural Operator (FNO) (LI et al., 2021) is one of the most widely adopted instance of this idea, it parameterizes an integral kernel in Fourier space, achieving speed-ups over other neural operators and is discretization-independent, the so-called zero-shot superesolution. Subsequent works explored alternative bases like wavelets (GUPTA et al., 2021), but FNO, was validated on many types of equations such as dispersive and soliton wave equations (PEI et al., 2022; ZHONG et al., 2023), was also caled to geophysical problems spanning regional ocean modeling, shallow-water emulation, and global weather forecasting (BONEV et al., 2023; LKHAGVASUREN et al., 2024; UCHIDA et al., 2025; PATEL et al., 2025; YU et al., 2025).

Following the idea for PINNs, The Physics-Informed Neural Operator (PINO) (LI et al., 2023a) combines operator learning with PDE-residual constraints, cutting the labelled-data requirement while preserving physical consistency. Applications to shallow-water systems (ROSOFSKY et al., 2023) confirmed that this hybrid regime inherits the generalization of FNO without its data hunger. Broad surveys of the field (TODO, 2023) identify these physics-data approaches working together as the most promising SciML direction, although highlighting turbulence modeling, scalability, and noisy-data robustness as open challenges.

A known limitation that cuts across all SciML architectures is the so-called spectral bias which is the tendency from gradient-based optimizers to learn low-frequency components of the solution faster than the high-frequency ones (RAHAMAN et al., 2019). Wang, Yu, et al. (2022) formalized this phenomenon through the lens of Neural Tangent Kernel (NTK) theory, showing that eigenvalue disparity between the PDE and initial-condition loss components drives the frequency imbalance during training. This pathology is especially consequential for dispersive wave problems, where solution energy is broadly distributed across wavenumbers; a recent systematic study (KHODAKARAMI et al., 2026) showed that second-order optimizers combined with spectrum-aware loss formulations substantially mitigate spectral bias.

As far as this review reveals, SciML methods have been applied to a broad class of shallow-water and dispersive wave equations. To the best of our knowledge, however, no study has systematically evaluated FNO, PINO, or PINNs on the parametric 1D Boussinesq system, where both the nonlinearity coefficient α and the dispersion coefficient β vary simultaneously. This gap motivates our work.

1.2 The Boussinesq System of Equations

The equations studied in this work originate from J. Boussinesq's 1872 effort to give a theoretical account of the solitary wave of translation observed by J. Scott Russell in a Scottish canal in 1834 (BOUSSINESQ, 1872). Russell had noticed that a hump of water, set in motion by a stopping barge, could travel for miles without changing shape, a phenomenon the linear wave theory of the time could not explain. Boussinesq showed that the stability of such a wave is the result of a precise balance between nonlinear steepening and frequency dispersion, two effects that are separately destabilizing but mutually compensating in the shallow-water, long-wave regime. The mathematical underpinnings of this balance, and of dispersive wave theory more broadly, are treated in depth by Whitham (1974) and Debnath (2012).

The modern two-field form of the system, for the free-surface elevation $\eta(x,t)$ and the depth-averaged horizontal velocity $u(x,t)$, was systematically derived and classified by Bona et al. (2002), who identified a four-parameter family of equivalent Boussinesq systems obtainable from the incompressible, irrotational Euler equations by asymptotic expansion in the shallowness and amplitude parameters. Earlier, Peregrine (1967) had already written down the standard form of these equations for water of variable depth, establishing the notation and scaling that remain in widespread use. What makes the Boussinesq system particularly suitable as a parametric testbed is that changing the nonlinearity and dispersion coefficients continuously deforms the solution from the near-linear, wave-packet regime to the strongly nonlinear soliton regime, spanning a rich range of behaviors within a single mathematical framework.

The specific one-dimensional form of the system used in this work, with its parameterization of the nonlinearity coefficient α and the dispersion coefficient β, was adopted from (MARTINS, 2023; MARTINS et al., 2024). In the dataset generated for this study α and β are held equal, defining a one-parameter family of problems whose difficulty grows with the common value: for small $\alpha = \beta$ the dynamics are nearly linear and the solution is well-captured by low-frequency modes, while for large $\alpha = \beta$ the strong coupling between nonlinearity and dispersion produces sharper, more energetically distributed wave profiles that are fundamentally harder to learn. This controlled variation is precisely what makes the setting informative for comparing SciML architectures.

Classical numerical methods for the Boussinesq system rely on pseudospectral and finite-difference discretizations that exploit the periodic or nearly periodic spatial structure of the waves. Pseudospectral schemes achieve spectral accuracy on the spatial derivatives and are therefore well suited to the dispersive character of the equations (ENGSIG-KARUP et al., 2012); this is the approach used to generate the reference dataset in this work. On the application side, operational coastal-engineering codes such as FUNWAVE (TODO, 2010) implement Boussinesq-type solvers for harbor and nearshore wave propagation, demonstrating the practical relevance of the model beyond the academic setting.

SciML methods have only recently been brought to bear on the Boussinesq equation. Zheng et al. (2023) applied parameter-integrated PINNs with phase-domain decomposition to reconstruct two-dimensional Boussinesq dynamics including phase transitions and rogue-wave formation, identifying spectral bias as a key obstacle at high nonlinearity. Wang et al. (2024) trained a physics-informed network to infer velocity and pressure fields from surface-elevation data alone on a Boussinesq model, demonstrating the inverse-problem capabilities of the approach. Complementary studies on related nonlinear wave equations (ZHANG et al., 2024; KUMAR et al., 2025) have confirmed that purely data-driven and purely physics-driven networks each display characteristic failure modes at strong nonlinearity. Taken together, these results suggest that a systematic, parametric evaluation of the full FNO/PINO/PINN family on the 1D Boussinesq system, which is the focus of this work, is both timely and necessary.

1.3 Research Goals

This work implements and evaluates three SciML methods on the parametric 1D Boussinesq system: Physics-Informed Neural Networks (PINNs), Fourier Neural Operators (FNOs), and Physics-Informed Neural Operators (PINOs). All results are measured against a pseudospectral reference solver, which sets a high-accuracy baseline. The study is guided by the following questions.

- Accuracy relative to the pseudospectral baseline: how closely can each SciML method reproduce the reference solutions as nonlinearity and dispersion grow? Where does each method start to fail, and how does its error scale with the difficulty of the regime?

- Data-driven, physics-informed, and hybrid training: how does the presence or absence of supervised reference data affect the accuracy and training behavior of each method? Does adding a physics residual term improve generalization when data are scarce, or does it introduce new difficulties?

- Spectral bias across training regimes: how does spectral bias manifest in each architecture, and to what extent does the inclusion of data or physics constraints alter the frequency at which errors concentrate during training?

Together, these questions aim to provide a clear picture of the strengths and limitations of each SciML approach for dispersive, nonlinear wave problems, and to offer practical guidance for choosing among them.

1.4 Work Structure

This work is written with the goal of sharing SciML for senior undergraduates in mathematics, physics, or engineering. We recommend the reader to have some knowledge of scientific computing and linear algebra. The remaining chapters are organized as follows.

- Chapter 2 develops the mathematical background. It covers the Boussinesq system and its analytical properties, the pseudospectral solver, and the theoretical foundations of MLP, PINN, FNO, and PINO, including a discussion of spectral bias.

- Chapter 3 describes the experimental methodology. It details the dataset generation procedure, the model architectures, the training setup, and the evaluation metrics used throughout.

- Chapter 4 presents the numerical results and discusses the performance of each method across parameter regimes and spatial resolutions.

- Chapter 5 summarizes the main findings and outlines directions for future work.

2 MATHEMATICAL BACKGROUND

This chapter introduces the mathematical foundations of the methods studied in this work. Section 2.1 presents the Boussinesq system and the limiting wave-equation regime. Section 2.2 develops the pseudospectral numerical method used to generate reference solutions. Sections 2.3 and 2.4 introduce, respectively, the neural network and neural operator architectures studied in this work.

2.1 The Boussinesq System

The Boussinesq system is a model for long nonlinear dispersive waves in shallow water, governing the coupled evolution of two scalar fields: the free surface elevation $\eta(x,t)$ and the depth-averaged horizontal velocity $u(x,t)$, both depending on position x and time t. In the one-dimensional, spatially periodic setting studied in this work, the full system reads:

$$\begin{cases} \eta_t + u_x + \alpha\,(\eta u)_x = 0, \\ u_t - \dfrac{\beta}{3}\,u_{xxt} + \eta_x + \alpha\,u u_x = 0, \\ \eta(x,0) = A\,\operatorname{sech}^2(x), \quad u(x,0) = 0, \quad x \in [-L, L], \end{cases} \tag{2.1}$$

where $\alpha \geq 0$ is the nonlinearity parameter, $\beta > 0$ is the dispersion parameter, $A > 0$ is the initial amplitude, $L > 0$ is the domain half-length, and $\operatorname{sech}(z) = 2/(e^z + e^{-z})$ is the hyperbolic secant. The initial condition places a solitary-wave-like crest at the center $x = 0$ with zero initial velocity. This nondimensional form follows equation (3.1) of Martins (2023) (see also Martins et al. (2024)).

We note that $uu_x = \frac{1}{2}(u^2)_x$, so the momentum equation in (2.1) can equivalently be written with the conservative flux $\frac{\alpha}{2}(u^2)_x$ in place of $\alpha\,uu_x$. Both forms appear in the implementations: the PINN residual uses uu_x directly via automatic differentiation, while the pseudospectral solver exploits $\frac{1}{2}(u^2)_x$ for the physical-space product, as detailed in Section 2.2.

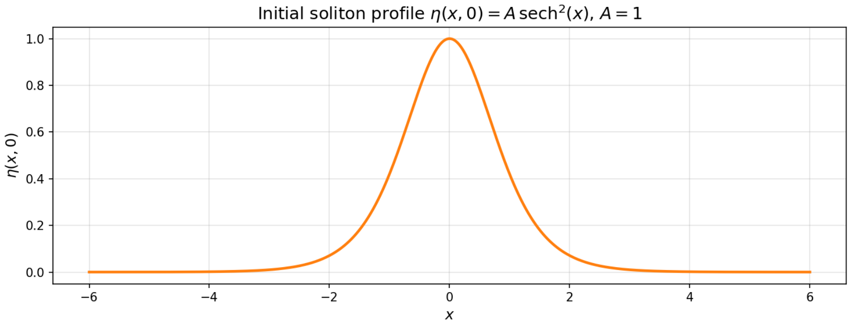

Figure 2.1 – Initial condition $\eta(x, 0) = A\,\mathrm{sech}^2(x)$ for $A = 1$.

Source: The author (2026).

2.1.1 Recovering the Wave Equation

In the doubly linear limit $\alpha = \beta = 0$, all nonlinear and dispersive terms vanish and the Boussinesq system reduces to the symmetric hyperbolic system:

$$\eta_t + u_x = 0, \tag{2.2}$$
$$u_t + \eta_x = 0. \tag{2.3}$$

To obtain a single scalar equation, differentiate (2.2) with respect to t:

$$\eta_{tt} + u_{xt} = 0. \tag{2.4}$$

Differentiate (2.3) with respect to x:

$$u_{tx} + \eta_{xx} = 0. \tag{2.5}$$

Since $u_{xt} = u_{tx}$, subtracting the second from the first yields the wave equation for η:

$$\eta_{tt} = \eta_{xx}, \tag{2.6}$$

and the reverse procedure gives the same equation for u.

D'Alembert's formula (DEBNATH, 2012) provides the general solution to (2.6). Given initial data

$$\eta(x, 0) = f(x), \qquad \eta_t(x, 0) = g(x), \tag{2.7}$$

the solution is:

$$\eta(x, t) = \frac{f(x + t) + f(x - t)}{2} + \frac{1}{2}\int_{x-t}^{x+t} g(s)\, ds. \tag{2.8}$$

We now apply this to the initial conditions (2.1). From (2.2), $\eta_t(x, 0) = -u_x(x, 0) =$

0, so:

$$f(x) = A\,\mathrm{sech}^2(x), \qquad g(x) = 0. \tag{2.9}$$

Since $g \equiv 0$, the integral in (2.8) vanishes and the surface elevation evolves as:

$$\eta(x,t) = \frac{A}{2}\big[\mathrm{sech}^2(x+t) + \mathrm{sech}^2(x-t)\big]. \tag{2.10}$$

For u, from (2.3) we have $u_t(x,0) = -\eta_x(x,0)$, giving:

$$f(x) = 0, \qquad g(x) = -\eta_x(x,0). \tag{2.11}$$

The $f = 0$ term vanishes in (2.8), leaving:

$$u(x,t) = \frac{1}{2}\int_{x-t}^{x+t} \big(-\eta_x(s,0)\big)\,ds = -\frac{1}{2}\big[\eta(x+t,0) - \eta(x-t,0)\big]. \tag{2.12}$$

Substituting $\eta(x,0) = A\,\mathrm{sech}^2(x)$:

$$u(x,t) = \frac{A}{2}\big[\mathrm{sech}^2(x-t) - \mathrm{sech}^2(x+t)\big]. \tag{2.13}$$

The initial bump of amplitude A splits into two counter-propagating pulses of amplitude $A/2$, traveling at speed ± 1 without changing shape. Any deviation from this wave-equation baseline in the full system (2.1) is attributable either to dispersion ($\beta > 0$) or to nonlinearity ($\alpha > 0$).

2.2 Pseudospectral Method

The pseudospectral method exploits two complementary representations of the solution: Fourier space, where derivatives become exact algebraic multiplications, and physical space, where pointwise nonlinear operations are inexpensive. By alternating between the two representations as needed, the method attains spectral accuracy in space while handling the nonlinear coupling efficiently.

2.2.1 Spatial Discretization and the Discrete Fourier Transform

The spatial domain $\Omega = [-L, L]$ of total length $2L$ is discretized into N_x equispaced points:

$$x_j = -L + j\,\Delta x, \qquad \Delta x = \frac{2L}{N_x}, \qquad j = 0, \ldots, N_x - 1. \tag{2.14}$$

The Discrete Fourier Transform (DFT) of a function f sampled at these points is (TREFETHEN, 2000):

$$\hat{f}(k) = \sum_{j=0}^{N_x-1} f(x_j)\, e^{-2\pi ikj/N_x}, \qquad k = -\tfrac{N_x}{2}, \ldots, \tfrac{N_x}{2} - 1, \tag{2.15}$$

with its inverse recovering the physical values:

$$f(x_j) = \frac{1}{N_x} \sum_k \hat{f}(k)\, e^{2\pi ikj/N_x}. \tag{2.16}$$

The fundamental identity of the DFT with respect to differentiation is that, for a periodic function, the spatial derivative transforms into a pointwise multiplication (TREFETHEN, 2000; ENGSIG-KARUP et al., 2012):

$$\widehat{(f_x)}(k) = i\frac{\pi k}{L}\, \hat{f}(k), \quad \widehat{(f_{xx})}(k) = -\frac{\pi^2 k^2}{L^2}\, \hat{f}(k). \tag{2.17}$$

This is in contrast to finite-difference methods, where derivatives are approximated with truncation error; in the DFT, equation (2.17) is exact for periodic functions.

We define the wavenumber of mode k on $[-L, L]$ as:

$$\xi_k \;:=\; \frac{\pi k}{L}. \tag{2.18}$$

With this definition, the derivative identities read:

$$\widehat{(f_x)}(k) = i\, \xi_k\, \hat{f}(k), \qquad \widehat{(f_{xx})}(k) = -\xi_k^2\, \hat{f}(k). \tag{2.19}$$

2.2.2 Spectral Formulation of the Boussinesq System

To bring the Boussinesq system (2.1) into the Fourier domain (MARTINS, 2023), we apply the DFT to each equation and use the derivative identities (2.19). The first equation, treating $\widehat{(\eta u)}(k,t)$ as a given nonlinear term, becomes:

$$\hat{\eta}_t(k,t) = -i\, \xi_k\, \hat{u}(k,t) - \alpha\, i\, \xi_k\, \widehat{(\eta u)}(k,t). \tag{2.20}$$

The second equation requires more care because of the u_{xxt} term. Since spatial derivatives commute with the DFT:

$$\widehat{(u_{xxt})}(k,t) = -\xi_k^2\, \hat{u}_t(k,t). \tag{2.21}$$

Substituting into the DFT of the momentum equation in (2.1):

$$\hat{u}_t(k,t) + \frac{\beta}{3}\,\xi_k^2\,\hat{u}_t(k,t) + i\,\xi_k\,\hat{\eta}(k,t) + \alpha\,i\,\xi_k\,\widehat{\left(\tfrac{1}{2}u^2\right)}(k,t) = 0. \tag{2.22}$$

Collecting the $\hat{u}_t(k,t)$ terms on the left and solving algebraically:

$$\hat{u}_t(k,t) = \frac{-i\,\xi_k\,\hat{\eta}(k,t) - \alpha\,i\,\xi_k\,\widehat{\left(\tfrac{1}{2}u^2\right)}(k,t)}{1 + \dfrac{\beta\,\xi_k^2}{3}}. \tag{2.23}$$

The operator $1-\frac{\beta}{3}\frac{\partial^2}{\partial x^2}$, which requires a linear solve in physical space, becomes the diagonal scalar $1+\frac{\beta\,\xi_k^2}{3}$ in Fourier space, reducing its inversion to a single division per mode.

Equations (2.20) and (2.23) can be written jointly as a first-order ODE system in time for each fixed wavenumber $k \in \mathbb{Z}$:

$$\begin{cases} \hat{\eta}_t(k,t) = -i\,\xi_k\,\hat{u}(k,t) - \alpha\,i\,\xi_k\,\widehat{(\eta u)}(k,t), \\[2ex] \hat{u}_t(k,t) = \dfrac{-i\,\xi_k\,\hat{\eta}(k,t) - \alpha\,i\,\xi_k\,\widehat{\left(\tfrac{1}{2}u^2\right)}(k,t)}{1 + \dfrac{\beta\,\xi_k^2}{3}}. \end{cases} \tag{2.24}$$

The PDE system (2.1), two equations in (x,t), is thus equivalent to an infinite family of two-dimensional ODE systems (2.24), one for each $k \in \mathbb{Z}$, coupled through the nonlinear terms.

The nonlinear terms $\widehat{(\eta u)}(k,t)$ and $\widehat{(\frac{1}{2}u^2)}(k,t)$ cannot be computed directly in Fourier space without a convolution sum, which costs $O(N_x^2)$. The pseudospectral approach avoids this by computing products in physical space. For $\widehat{(\eta u)}(k,t)$, the procedure is:

1. recover $\eta(x_j)$ and $u(x_j)$ by applying the inverse DFT (2.16);
2. compute the product $(\eta u)_j = \eta(x_j)\cdot u(x_j)$ pointwise on the grid;
3. apply the DFT (2.15) to obtain $\widehat{(\eta u)}(k,t)$.

Similarly, $\widehat{(\frac{1}{2}u^2)}(k,t)$ is obtained by the same three-step procedure with $u^2(x_j) = u(x_j)^2$, where $u(x_j) = \mathrm{IDFT}(\hat{u})(x_j)$; and $(\eta u)(x_j) = \mathrm{IDFT}(\hat{\eta})(x_j)\cdot\mathrm{IDFT}(\hat{u})(x_j)$. Since each DFT/IDFT costs $O(N_x \log N_x)$ via the Fast Fourier Transform (FFT), this pseudospectral strategy assembles each nonlinear term in $O(N_x \log N_x)$ rather than $O(N_x^2)$.

2.2.3 Time Evolution via Runge-Kutta 4

We define the pseudospectral right-hand side operator F as the map

$$F(\hat{\eta},\hat{u})\,(k) \;=\; \begin{pmatrix} -i\,\xi_k\,\hat{u}(k) - \alpha\, i\,\xi_k\,\widehat{(\eta u)}(k) \\[2ex] \dfrac{-i\,\xi_k\,\hat{\eta}(k) - \alpha\, i\,\xi_k\,\widehat{\big(\tfrac{1}{2}u^2\big)}(k)}{1+\dfrac{\beta\,\xi_k^2}{3}} \end{pmatrix}, \tag{2.25}$$

so that $F(\hat{\eta},\hat{u}) = \big(\hat{\eta}_t,\,\hat{u}_t\big)$ as given by equations (2.20)–(2.23), with nonlinear terms assembled by the pseudospectral procedure above. Time evolution uses the classical fourth-order Runge-Kutta (RK4) scheme (SÜLI et al., 2003; BURDEN et al., 2011). Given the spectral state $y_n = (\hat{\eta}^n, \hat{u}^n)$ at time t_n, the four stage evaluations $\kappa_1, \kappa_2, \kappa_3, \kappa_4$ and the update to $t_{n+1} = t_n + \Delta t$ are:

$$\begin{aligned} \kappa_1 &= F(y_n), \\ \kappa_2 &= F\big(y_n + \tfrac{\Delta t}{2}\,\kappa_1\big)\,, \\ \kappa_3 &= F\big(y_n + \tfrac{\Delta t}{2}\,\kappa_2\big)\,, \\ \kappa_4 &= F(y_n + \Delta t\,\kappa_3)\,, \\ y_{n+1} &= y_n + \frac{\Delta t}{6}\,(\kappa_1 + 2\,\kappa_2 + 2\,\kappa_3 + \kappa_4). \end{aligned} \tag{2.26}$$

At each stage, the intermediate state is decoded to physical space to assemble the nonlinear products, then encoded back to Fourier space to evaluate the spectral derivatives. This split between physical-space products and Fourier-space derivatives is the defining operation of the pseudospectral method.

2.2.4 Training Dataset

The numerical experiments in this work vary only the PDE parameters (α,β) while keeping the initial condition fixed at (2.1). The models are therefore trained and evaluated on the dataset

$$\mathcal{S} = \big\{\big((\alpha_i,\beta_i),\,(\eta_i,u_i)\big)\big\}_{i=1}^{M}, \tag{2.27}$$

where (α_i,β_i) are the nonlinearity and dispersion parameters for the i-th sample, and (η_i,u_i) is the corresponding space-time solution obtained from the pseudospectral solver of Section 2.2 with those parameters and the shared initial condition (2.1).

2.3 Neural Networks

This section introduces the neural network architectures central to this work together with the theory that explains their training behavior. We begin with the Multilayer Perceptron as the building block, develop the Neural Tangent Kernel framework and the spectral bias it predicts for a network fitting data on a regular grid, and then turn to Physics-Informed Neural Networks, showing how the same kernel analysis extends to the physics-informed setting where spectral bias is a well-documented obstacle.

2.3.1 Multilayer Perceptrons

In this work, we focus on Multilayer Perceptrons (MLPs), popularized by Rumelhart et al. (1986). These networks are compositions of affine transformations parametrized by tunable weights and biases, separated by nonlinear activations, where an optimization process fits the network to a dataset.

Given a labeled dataset $\{(x_i, u_i)\}_{i=1}^{N}$, the objective is to fit the function $u_\theta(x)$ by minimizing a loss function, here the Mean Squared Error (MSE). With $\|\cdot\|$ denoting the Euclidean norm, the unconstrained optimization problem is:

$$\mathcal{L}_{\text{MLP}}(\theta) = \frac{1}{N}\sum_{i=1}^{N} \|u_\theta(x_i) - u_i\|^2. \tag{2.28}$$

The MLP $u_\theta : \mathbb{R}^{d_{\text{in}}} \to \mathbb{R}^{d_{\text{out}}}$ is defined by the composition:

$$h^{(0)} = x \in \mathbb{R}^{d_{\text{in}}}, \tag{2.29}$$

$$h^{(l)} = \sigma\left(W^{(l)} h^{(l-1)} + b^{(l)}\right) \in \mathbb{R}^{N_{\text{MLP}}}, \quad l = 1, \dots, L_{\text{MLP}} - 1, \tag{2.30}$$

$$u_\theta(x) = W^{(L_{\text{MLP}})} h^{(L_{\text{MLP}}-1)} + b^{(L_{\text{MLP}})} \in \mathbb{R}^{d_{\text{out}}}. \tag{2.31}$$

where:

- $L_{\text{MLP}} \in \mathbb{N}$ is the number of layers and $N_{\text{MLP}} \in \mathbb{N}$ the number of neurons per hidden layer;
- $W^{(l)} \in \mathbb{R}^{N_{\text{MLP}} \times N_{\text{MLP}}}$ and $b^{(l)} \in \mathbb{R}^{N_{\text{MLP}}}$ are the weight matrix and bias vector at layer l, with $W^{(1)} \in \mathbb{R}^{d_{\text{in}} \times N_{\text{MLP}}}$ and $W^{(L_{\text{MLP}})} \in \mathbb{R}^{N_{\text{MLP}} \times d_{\text{out}}}$ adapted at the boundaries;
- $\sigma : \mathbb{R} \to \mathbb{R}$ is a sigmoidal activation function applied elementwise, satisfying $\sigma(t) \to 1$ as $t \to +\infty$ and $\sigma(t) \to 0$ as $t \to -\infty$;
- $\theta = \{W^{(l)}, b^{(l)}\}_{l=1}^{L_{\text{MLP}}}$ denotes all trainable parameters.

If the activation function is sigmoidal, MLPs are universal approximators: they can approximate any continuous function on a compact domain to arbitrary precision (CYBENKO, 1989).

In Figure 2.2, we show a common depiction of the neural network. Here the input vector is the pair $(x,t) \in \mathbb{R}^2$ and the output is $(\eta_\theta(x,t), u_\theta(x,t)) \in \mathbb{R}^2$.

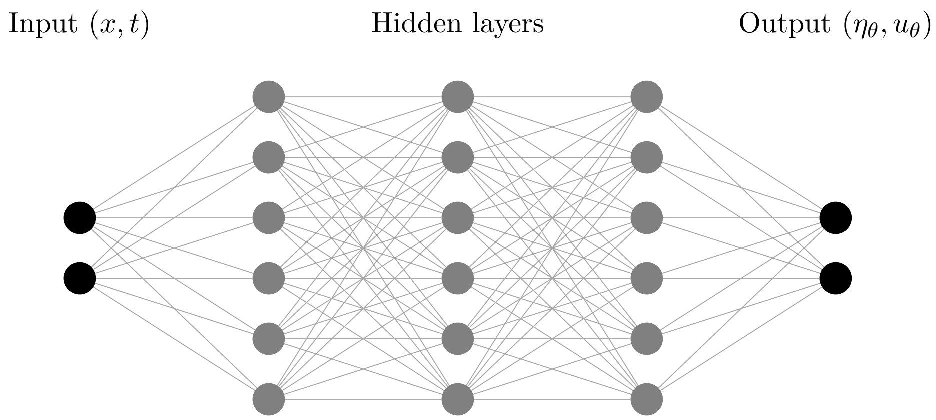

Figure 2.2 – Example of MLP architecture of input (x,t), with $L_{\mathrm{MLP}} = 3$ hidden layers and $N_{\mathrm{MLP}} = 6$ neurons per layer.

Source: The author (2026).

2.3.2 Gradient Flow

To understand how an MLP minimizes its data-fitting loss (2.28), consider gradient descent with learning rate $\mu > 0$. Starting from an initialization $\theta^{(0)}$, each iteration updates the parameters by

$$\theta^{(k+1)} = \theta^{(k)} - \mu \, \nabla_\theta \mathcal{L}(\theta^{(k)}), \qquad k = 0, 1, 2, \dots, \tag{2.32}$$

where $\theta^{(k)}$ denotes the parameter vector after k steps. Rearranging (2.32) and dividing by the step size μ isolates the per-step increment as a difference quotient:

$$\frac{\theta^{(k+1)} - \theta^{(k)}}{\mu} = -\nabla_\theta \mathcal{L}(\theta^{(k)}). \tag{2.33}$$

To pass from this discrete recursion to a continuous trajectory, assign to each iterate a training-time stamp

$$\tau^{(k)} = k\mu, \qquad \text{so that} \qquad \tau^{(k+1)} - \tau^{(k)} = \mu, \tag{2.34}$$

spacing successive iterates by μ along a time axis τ. We then regard the discrete sequence $\{\theta^{(k)}\}_{k=0}^{\infty}$ as samples of a smooth curve $\theta(\tau)$ evaluated at these instants:

$$\theta^{(k)} = \theta(\tau^{(k)}), \qquad \theta^{(k+1)} = \theta(\tau^{(k+1)}) = \theta(\tau^{(k)} + \mu). \tag{2.35}$$

Substituting (2.35) into the left-hand side of (2.33) recasts it as a forward finite-difference quotient of θ in τ:

$$\frac{\theta(\tau^{(k)} + \mu) - \theta(\tau^{(k)})}{\mu} = -\nabla_\theta \mathcal{L}\big(\theta(\tau^{(k)})\big). \tag{2.36}$$

As the step size $\mu \to 0$, the spacing in (2.34) shrinks to zero, the discrete stamps $\tau^{(k)}$ fill out a continuous variable τ, and the left-hand side of (2.36) converges to the derivative $d\theta/d\tau$ by definition of the derivative. The recursion thus becomes the gradient flow ODE:

$$\frac{d\theta}{d\tau} = -\nabla_\theta \mathcal{L}(\theta), \qquad \theta(0) = \theta^{(0)}. \tag{2.37}$$

The gradient flow (2.37) is a central object in optimization, and it is a key insight that many standard first-order methods can be recovered as different time-discretization schemes of this same continuous trajectory. The discrete update (2.32) is precisely the forward (explicit) Euler discretization of (2.37) with step size μ, so gradient descent is just one numerical integrator for the flow; applying instead the backward (implicit) Euler scheme recovers the proximal gradient method (PARIKH et al., 2013). Reading optimizers as discretizations of (2.37) thus organizes them into a single family, and it is the continuous flow that we analyze in what follows.

2.3.3 Neural Tangent Kernel

Consider the MLP fit to the dataset $\{(x_i, u_i)\}_{i=1}^{N}$ of Section 2.3.1. Group the pointwise fitting errors into a single error vector:

$$e(\theta) = \begin{bmatrix} u_\theta(x_1) - u_1 \\ \vdots \\ u_\theta(x_N) - u_N \end{bmatrix} \in \mathbb{R}^N. \tag{2.38}$$

Up to the constant factor $1/N$, the loss (2.28) is the squared norm of this vector, which we use in the form

$$\mathcal{L}(\theta) = \frac{1}{2}\,\|e(\theta)\|^2 = \frac{1}{2}\sum_{i=1}^{N} e_i(\theta)^2. \tag{2.39}$$

Define the Jacobian of the error vector with respect to the parameters:

$$J(\theta) = \frac{\partial e}{\partial \theta} \in \mathbb{R}^{N \times N_\theta}, \qquad J_{ij} = \frac{\partial e_i}{\partial \theta_j}, \quad i = 1, \dots, N, \ j = 1, \dots, N_\theta, \tag{2.40}$$

where N_θ is the total number of trainable parameters. Each row $J_i = (J_{i1}, \dots, J_{iN_\theta}) \in \mathbb{R}^{N_\theta}$ is the parameter-gradient of the i-th residual. Since $\mathcal{L} = \frac{1}{2}\sum_i e_i^2$, the j-th component of the loss gradient is:

$$\frac{\partial \mathcal{L}}{\partial \theta_j} = \sum_{i=1}^{N} e_i(\theta)\, \frac{\partial e_i}{\partial \theta_j} = \sum_{i=1}^{N} J_{ij}\, e_i(\theta) = \big(J(\theta)^T e(\theta)\big)_j. \tag{2.41}$$

Collecting all $j = 1, \dots, N_\theta$ into a column vector:

$$\nabla_\theta \mathcal{L} = J(\theta)^T\, e(\theta) \in \mathbb{R}^{N_\theta}. \tag{2.42}$$

Substituting (2.42) into the gradient flow (2.37):

$$\frac{d\theta}{d\tau} = -J(\theta)^T\, e(\tau). \tag{2.43}$$

To find how the error vector itself evolves, differentiate each component $e_i(\theta(\tau))$ with respect to τ by the chain rule:

$$\frac{de_i}{d\tau} = \sum_{j=1}^{N_\theta} \frac{\partial e_i}{\partial \theta_j}\, \frac{d\theta_j}{d\tau} = J_i(\theta) \cdot \frac{d\theta}{d\tau}. \tag{2.44}$$

Stacking this identity for all $i = 1, \dots, N$ into a matrix equation:

$$\frac{de}{d\tau} = J(\theta)\, \frac{d\theta}{d\tau}. \tag{2.45}$$

Substituting (2.43):

$$\frac{de}{d\tau} = J(\theta)\big(-J(\theta)^T e(\tau)\big) = -\underbrace{J(\theta)\, J(\theta)^T}_{K(\theta) \,\in\, \mathbb{R}^{N \times N}}\, e(\tau). \tag{2.46}$$

The matrix $K(\theta) = J(\theta)J(\theta)^T$ is the empirical Neural Tangent Kernel (NTK) (JACOT et al., 2018). It is symmetric positive semi-definite by construction: for any $v \in \mathbb{R}^N$,

$$v^T K(\theta)\, v = v^T J(\theta)\, J(\theta)^T v = \big\| J(\theta)^T v \big\|^2 \geq 0. \tag{2.47}$$

The (i, j) entry of $K(\theta)$ follows directly from expanding the matrix product $J(\theta)J(\theta)^T$:

$$K_{ij}(\theta) = \sum_{l=1}^{N_\theta} J_{il}(\theta)\, J_{jl}(\theta) = \sum_{l=1}^{N_\theta} \frac{\partial e_i}{\partial \theta_l}(\theta)\, \frac{\partial e_j}{\partial \theta_l}(\theta) = \nabla_\theta e_i(\theta)^T\, \nabla_\theta e_j(\theta). \tag{2.48}$$

This is the inner product in $\mathbb{R}^{N_\theta}$ of the parameter gradients of the network output at the i-th and j-th training points, where $e_i = u_\theta(x_i) - u_i$. Hence $K(\theta) \in \mathbb{R}^{N\times N}$ is the Gram matrix of the network's sensitivities across the whole dataset, and equation (2.46) shows that it alone governs how the fitting error evolves under gradient flow.

2.3.4 Spectral Bias

The spectral bias (or frequency principle) is the empirical observation that neural networks trained by gradient-based methods reduce the low-frequency components of the error faster than the high-frequency ones (RAHAMAN et al., 2019). This section derives that phenomenon directly from the NTK dynamics established above. When the training points lie on a regular grid and the target function is periodic, the error admits a discrete Fourier representation, and one can ask how its frequency content is approximated along the training iterations. We first establish the decay in the eigenbasis of the NTK and then, by transforming to Fourier space, show that it is precisely the low frequencies that are resolved first.

For a network with a sufficiently large number of parameters, the Jacobian $J(\theta)$ changes negligibly during training (JACOT et al., 2018; WANG; YU, et al., 2022). Writing $J_0 = J(\theta(0))$ and $K_0 = J_0 J_0^T$, the approximation $K(\theta(\tau)) \approx K_0$ turns equation (2.46) into a linear ODE with constant coefficients:

$$\frac{de}{d\tau} = -K_0\, e(\tau), \qquad e(0) = e_0. \tag{2.49}$$

Written component-wise, the i-th row of this system is

$$\frac{de_i}{d\tau} = -\sum_{j=1}^{N} K_0^{ij}\, e_j(\tau), \qquad i = 1, \dots, N, \tag{2.50}$$

where $K_0^{ij} = \nabla_{\theta_0} e_i^T\, \nabla_{\theta_0} e_j$ is the inner product, at initialization, of the parameter gradients of the network output at training points x_i and x_j. The rate of change of the error at point i is therefore a weighted sum of all current errors, with weights given by these pairwise kernel values. This coupling is what makes the geometry of the sample grid and the architecture of the network jointly determine the convergence of every component.

Since $K_0 \in \mathbb{R}^{N\times N}$ is symmetric and positive semi-definite, the spectral theorem guarantees a decomposition

$$K_0 = V\,\Lambda\,V^T, \tag{2.51}$$

where $V = [v_1 \mid \cdots \mid v_N]$ is an orthogonal matrix ($V^T V = I$) and

$$\Lambda = \operatorname{diag}(\lambda_1, \ldots, \lambda_N), \qquad \lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_N \geq 0. \tag{2.52}$$

Introduce the change of coordinates

$$c(\tau) = V^T e(\tau), \tag{2.53}$$

so that, since V is orthogonal, $e(\tau) = V c(\tau)$. Differentiating (2.53) with respect to τ and substituting the ODE (2.49):

$$\frac{dc}{d\tau} = V^T \frac{de}{d\tau} = -V^T K_0\, e(\tau). \tag{2.54}$$

Inserting $K_0 = V\Lambda V^T$ and $e(\tau) = V c(\tau)$:

$$\frac{dc}{d\tau} = -V^T\,(V\Lambda V^T)\,(V c(\tau)) = -(V^T V)\,\Lambda\,(V^T V)\,c(\tau) = -\Lambda\,c(\tau), \tag{2.55}$$

using $V^T V = I$ at each step. Since Λ is diagonal, this decouples into N independent scalar equations:

$$\frac{dc_k}{d\tau} = -\lambda_k\,c_k(\tau), \qquad k = 1, \ldots, N. \tag{2.56}$$

Each equation in (2.56) is a first-order linear ODE with constant coefficient (STROGATZ, 2018). Separating the variables and integrating both sides from the initial state at $\tau = 0$ up to time τ, with s and r as integration variables for the amplitude and the time,

$$\int_{c_k(0)}^{c_k(\tau)} \frac{ds}{s} = -\lambda_k \int_0^\tau dr \qquad \Longrightarrow \qquad \ln\frac{|c_k(\tau)|}{|c_k(0)|} = -\lambda_k\,\tau. \tag{2.57}$$

Exponentiating both sides yields the solution

$$c_k(\tau) = c_k(0)\,e^{-\lambda_k \tau}, \tag{2.58}$$

where $c_k(0) = v_k^T e_0$ is the projection of the initial error e_0 onto the k-th eigenvector of K_0. For $\lambda_k = 0$ the equation gives $c_k(\tau) = c_k(0)$ for all τ: that eigendirection never decays.

Returning to the original coordinates via $e(\tau) = V c(\tau)$:

$$e(\tau) = \sum_{k=1}^{N} \left(v_k^T e_0\right) e^{-\lambda_k \tau}\,v_k. \tag{2.59}$$

The error decomposes into N independent eigendirections. Each eigenvector v_k carries initial amplitude $v_k^T e_0$ and decays exponentially at rate λ_k: eigenvectors with large λ_k are suppressed quickly, while those with small λ_k persist throughout training. This per-eigendirection decay, each at the rate set by its eigenvalue, is the rigorous statement of spectral bias established by Cao et al. (2021); the same spectral ordering governs generalization as the training set grows (BORDELON et al., 2020).

Equation (2.59) is the continuous flow; gradient descent takes finite steps, and its error evolution inherits the same eigenstructure in discrete form. Linearizing the update (2.32) about the frozen Jacobian J_0, a single step moves the parameters by $\theta^{(n+1)} - \theta^{(n)} = -\mu\, J_0^T e^{(n)}$, and to first order the error responds as $e^{(n+1)} - e^{(n)} = J_0\,(\theta^{(n+1)} - \theta^{(n)})$. Combining the two,

$$e^{(n+1)} = e^{(n)} - \mu\, J_0 J_0^T e^{(n)} = (I - \mu K_0)\, e^{(n)}. \tag{2.60}$$

Iterating from $e^{(0)} = e_0$ and diagonalizing with $K_0 = V \Lambda V^T$,

$$e^{(n)} = (I - \mu K_0)^n e_0 = \sum_{k=1}^{N} \big(v_k^T e_0\big)\, (1 - \mu\lambda_k)^n\, v_k. \tag{2.61}$$

This is the forward-Euler counterpart of (2.59): each eigendirection is multiplied by $(1 - \mu\lambda_k)^n$ rather than $e^{-\lambda_k \tau}$, and the two coincide as $\mu \to 0$ with $\tau = n\mu$, since $(1 - \mu\lambda_k)^n \to e^{-\lambda_k\tau}$. The discrete mode contracts only when $|1 - \mu\lambda_k| < 1$, that is $\mu < 2/\lambda_k$, so the largest eigenvalue caps the stable learning rate at $\mu < 2/\lambda_{\max}$.

From (2.58), the amplitude of the k-th eigenvector at time τ satisfies $|c_k(\tau)| = |v_k^T e_0|\, e^{-\lambda_k \tau}$. To reduce this amplitude below a target precision $\varepsilon > 0$:

$$|v_k^T e_0|\, e^{-\lambda_k\tau} < \varepsilon \implies e^{-\lambda_k\tau} < \frac{\varepsilon}{|v_k^T e_0|} \implies -\lambda_k\, \tau < \ln \frac{\varepsilon}{|v_k^T e_0|} \implies \tau > \frac{1}{\lambda_k} \ln \frac{|v_k^T e_0|}{\varepsilon}. \tag{2.62}$$

The minimum training time to resolve the k-th eigenvector to precision ε is therefore

$$\tau_k = \frac{1}{\lambda_k} \ln \frac{|v_k^T e_0|}{\varepsilon}. \tag{2.63}$$

Since $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_N \geq 0$, equation (2.63) gives $\tau_1 \leq \tau_2 \leq \cdots \leq \tau_N$: higher-indexed eigenvectors require more training time. Arora et al. (2019) reached similar conclusions from a discrete-time analysis of gradient descent, showing that the convergence speed of each eigenvector is governed by its eigenvalue and by the projection of the target onto that eigenvector.

The decomposition (2.59) is expressed in the eigenbasis $\{v_k\}_{k=1}^N$ of the NTK, which need not coincide with any physically meaningful basis. When the N training points lie on a regular grid and the target is periodic, however, the natural basis is the Fourier one, and recasting the dynamics there shows that the decay is genuinely organized by frequency. Let $F \in \mathbb{C}^{N\times N}$ be the Fourier matrix, that is, the discrete Fourier transform (DFT) matrix with entries

$$F_{jl} = \frac{1}{\sqrt{N}}\,\omega^{jl}, \qquad \omega = e^{-2\pi \mathrm{i}/N}, \qquad j, l = 0, \ldots, N-1. \tag{2.64}$$

With this normalization F is unitary, $F^*F = I$, so that $F^{-1} = F^*$, where F^* is the conjugate transpose (GOLUB et al., 2013; STRANG, 2006). Applying F to the error vector yields its spectrum, with inverse transform F^{-1}:

$$\hat{e}(\tau) = F\,e(\tau), \qquad e(\tau) = F^{-1}\hat{e}(\tau) = F^*\hat{e}(\tau). \tag{2.65}$$

In particular $\hat{e}_0 = Fe_0$ is the spectrum of the initial error, the difference between the spectra of the initial network output and the target. Since F is constant in τ, applying it to the linearized dynamics (2.49) carries the error evolution into the frequency domain:

$$\frac{d\hat{e}}{d\tau} = F\,\frac{de}{d\tau} = -FK_0\,e(\tau) = -FK_0F^{-1}\,\hat{e}(\tau). \tag{2.66}$$

Inserting the eigendecomposition $K_0 = V\Lambda V^T$ from (2.51),

$$FK_0F^{-1} = FV\,\Lambda\,V^TF^{-1} = (FV)\,\Lambda\,(FV)^* = (FV)\,\Lambda\,(FV)^{-1}, \tag{2.67}$$

where $V^TF^{-1} = V^*F^* = (FV)^*$ uses that V is real orthogonal, and $(FV)^* = (FV)^{-1}$ uses that a product of unitary matrices is unitary. The columns of FV, namely the Fourier transforms $\{Fv_k\}_{k=1}^N$ of the NTK eigenvectors, therefore diagonalize the frequency-domain dynamics. Defining the modal amplitudes $\tilde{c}(\tau) = (FV)^{-1}\hat{e}(\tau)$ and repeating the steps (2.53)–(2.58) gives

$$\frac{d\tilde{c}_k}{d\tau} = -\lambda_k\,\tilde{c}_k(\tau) \qquad \Longrightarrow \qquad \tilde{c}_k(\tau) = \tilde{c}_k(0)\,e^{-\lambda_k\tau}. \tag{2.68}$$

These are the same amplitudes as before: $\tilde{c} = (FV)^{-1}Fe = V^{-1}e = V^Te = c$, the projection of the error onto the NTK eigenvectors, now read in Fourier space. Transforming back through (2.65) yields the evolution of the error spectrum,

$$\hat{e}(\tau) = \sum_{k=1}^{N}\big[(Fv_k)^*\hat{e}_0\big]\,e^{-\lambda_k\tau}\,(Fv_k). \tag{2.69}$$

Equation (2.69) is the spectral-bias statement in the literal sense of the frequency spectrum: the error spectrum is a superposition of the transformed eigenvectors Fv_k, each damped at its own rate λ_k. The frequencies resolved first are those carried by the eigenvectors with the largest eigenvalues; empirically these eigenvectors are smooth and concentrated in the low Fourier modes (RAHAMAN et al., 2019; XU et al., 2020), so the low-frequency content of the error decays first while the high-frequency content persists, exactly the behaviour observed in practice.

To close, note that when the network is initialized at small scale its output u_{θ_0} is negligible relative to the target, a common simplification in spectral-bias analyses. The initial error is then the target with opposite sign, and so is its spectrum:

$$\hat{e}_0 = F\big(u_{\theta_0} - u\big) = \hat{u}_{\theta_0} - \hat{u} \approx -\,\hat{u} = -\begin{bmatrix} \hat{u}(0) \\ \vdots \\ \hat{u}(N-1) \end{bmatrix}. \tag{2.70}$$

Substituting (2.70) into (2.69) casts the dynamics as a reconstruction of the target frequency by frequency: each Fourier mode of u is acquired at a rate set by the NTK eigenvalues, the smooth low-frequency content emerging first and the fine high-frequency detail last. This is exactly the regime anticipated at the start of the section — a periodic target on a regular grid — in which the quality of the approximation can be tracked mode by mode along the training iterations.

This last observation can be made quantitative. Taking the modulus of the modal amplitude (2.68) isolates the magnitude of the k-th component along training,

$$|\tilde{c}_k(\tau)| = |\tilde{c}_k(0)|\, e^{-\lambda_k \tau}, \qquad |\tilde{c}_k(0)| = |(Fv_k)^* \hat{e}_0| \approx |(Fv_k)^* \hat{u}|, \tag{2.71}$$

where the last estimate uses the small-scale initialization (2.70), so that $|(Fv_k)^*\hat{u}|$ measures how strongly the target overlaps with the k-th eigen-mode. Training, however, advances in discrete steps: by the construction of the gradient flow (2.34), the continuous time and the iteration count n are related by $\tau = n\mu$, with μ the learning rate. Substituting $\tau = n\mu$ into (2.71) and requiring the amplitude to fall below a target precision $\varepsilon > 0$,

$$|(Fv_k)^*\hat{u}|\, e^{-\lambda_k n\mu} < \varepsilon \implies n > \frac{1}{\mu\,\lambda_k} \ln \frac{|(Fv_k)^*\hat{u}|}{\varepsilon}, \tag{2.72}$$

gives an estimate of the number of gradient-descent iterations needed to resolve the k-th frequency:

$$n_k = \left\lceil \frac{1}{\mu\,\lambda_k} \ln \frac{|(Fv_k)^*\hat{u}|}{\varepsilon} \right\rceil. \tag{2.73}$$

2.3.5 Physics-Informed Neural Networks

Introduced by Raissi et al. (2017), Physics-Informed Neural Networks (PINNs) insert the PDE directly into the optimization process, forcing the neural network to satisfy the model equations and approximating the PDE solution.

Consider the general PDE Initial Value Problem:

$$\begin{cases} \mathcal{D}[u](x,t) = 0, \quad \forall (x,t) \in \Omega \times (0,T], \\ \quad u(x,0) = u_0(x), \quad x \in \Omega, \end{cases} \tag{2.74}$$

where $\mathcal{D}$ is a differential operator on u, $\Omega \times (0,T]$ is the spatio-temporal domain, and u_0 is the initial condition.

The PDE residual loss penalizes violation of the governing equation at collocation points $\{(x_i, t_i)\}_{i=1}^{N_{\mathcal{D}}} \subset \Omega \times (0,T]$:

$$\mathcal{L}_{\text{pde}}(\theta) = \frac{1}{N_{\mathcal{D}}} \sum_{i=1}^{N_{\mathcal{D}}} \left\| \mathcal{D}[u_\theta](x_i, t_i) \right\|^2 . \tag{2.75}$$

The initial condition is enforced at collocation points $\{x_i\}_{i=1}^{N_0} \subset \Omega$:

$$\mathcal{L}_{\text{ic}}(\theta) = \frac{1}{N_0} \sum_{i=1}^{N_0} \left\| u_\theta(x_i, 0) - u_0(x_i) \right\|^2 . \tag{2.76}$$

The combined PINN loss is:

$$\mathcal{L}_{\text{PINN}}(\theta) = \mathcal{L}_{\text{pde}}(\theta) + \mathcal{L}_{\text{ic}}(\theta). \tag{2.77}$$

Due to the compositional structure of MLPs, PINN implementations evaluate PDE residuals via Automatic Differentiation (AD) (LINNAINMAA, 1970; PASZKE et al., 2019), using ADAM (KINGMA et al., 2014) or L-BFGS (NOCEDAL, 1980) as the optimizer.

Application to the Boussinesq Dataset

For the Boussinesq system, a separate PINN is trained for each sample $(\alpha_i, \beta_i) \in \mathcal{S}$ (2.27). The differential operator $\mathcal{D}$ in the generic problem (2.74) is instantiated as the pair of Boussinesq residuals with fixed parameters (α_i, β_i):

$$\mathcal{D}_1[u_\theta] = (\eta_\theta)_t + (u_\theta)_x + \alpha_i \left((\eta_\theta \, u_\theta)_x \right), \tag{2.78}$$

$$\mathcal{D}_2[u_\theta] = (u_\theta)_t - \tfrac{\beta_i}{3} (u_\theta)_{xxt} + (\eta_\theta)_x + \alpha_i \left(u_\theta (u_\theta)_x \right). \tag{2.79}$$

The combined loss (2.77) for sample i then reads:

$$\begin{aligned}\mathcal{L}^{i}_{\text{PINN}}(\theta) = {} & \frac{1}{N_{\mathcal{D}}}\sum_{l=1}^{N_{\mathcal{D}}}\Big[\mathcal{D}_1[u_\theta](x_l,t_l)^2 + \mathcal{D}_2[u_\theta](x_l,t_l)^2\Big] \\ & + \frac{1}{N_0}\sum_{l=1}^{N_0}\Big[(\eta_\theta(x_l,0)-\eta_0(x_l))^2 + (u_\theta(x_l,0)-u_0(x_l))^2\Big],\end{aligned} \tag{2.80}$$

where η_0 and u_0 are given by the shared initial condition (2.1). Because the parameters (α_i, β_i) enter the residuals explicitly, each PINN encodes knowledge of a single sample and must be retrained for every new parameter pair.

2.3.6 Spectral Bias in Physics-Informed Neural Networks

The Neural Tangent Kernel analysis of Sections 2.3.3–2.3.4 was carried out for an MLP fitting data on a uniform grid, the setting in which the kernel is approximately circulant and its eigenvectors are genuine Fourier modes. A PINN does not fit data directly: its loss (2.77) combines an initial-condition term with a PDE-residual term. The same machinery nevertheless applies once both constraints are stacked into a single error vector, now indexed over the N_r residual points in place of the N data points of the MLP,

$$e(\theta) = \begin{bmatrix} u_\theta(x_1,0) - u_0(x_1) \\ \vdots \\ u_\theta(x_{N_0},0) - u_0(x_{N_0}) \\ \mathcal{D}[u_\theta](x_1,t_1) \\ \vdots \\ \mathcal{D}[u_\theta](x_{N_{\mathcal{D}}},t_{N_{\mathcal{D}}}) \end{bmatrix} \in \mathbb{R}^{N_r}, \qquad N_r = N_0 + N_{\mathcal{D}}, \tag{2.81}$$

so that, up to normalization, $\mathcal{L}_{\text{PINN}}$ reduces to the same quadratic form $\frac{1}{2}\|e\|^2$ analyzed for the MLP. With this definition the gradient-flow derivation leading to (2.46) carries over verbatim, with the Jacobian now also containing the parameter gradients of the PDE residuals, and the error still obeys the linear dynamics $\frac{de}{d\tau} = -K_0\, e$, with convergence of each eigenvector governed by its eigenvalue exactly as in (2.63) (WANG; YU, et al., 2022).

This prediction is borne out empirically. Spectral bias is among the most consistently reported obstacles in PINN training: networks fit smooth, low-frequency solutions readily but stall on sharp gradients and high-frequency content (RAHAMAN et al., 2019; XU et al., 2020; WANG; TENG, et al., 2021), and the same pathology underlies several documented PINN failure modes (KRISHNAPRIYAN et al., 2021). Wang, Yu, et al. (2022) traced the effect directly to the NTK, showing that its spectrum is concentrated at low eigenvalues, so that high-frequency PDE residuals are the last to be minimized; for the Boussinesq system specifically, Wang et al. (2024) report the same difficulty in resolving the finer wave structure. A range of remedies target precisely this imbalance, including adaptive loss weighting (WANG; TENG, et al., 2021), Fourier feature embeddings, and domain decomposition. Understanding the spectral bias is therefore essential background for the neural-operator methods studied in the remainder of this work, whose ability to capture the high-frequency content of the Boussinesq solutions is examined directly in the numerical results of Chapter 4.

2.4 Neural Operators

Neural operators generalize neural networks from learning finite-dimensional mappings to learning operators between infinite-dimensional function spaces. This section introduces the general framework, then the Fourier Neural Operator (FNO) as the principal architecture, and finally the Physics-Informed Neural Operator (PINO) as the physics-constrained extension.

2.4.1 General Definition

Let $D \subset \mathbb{R}^d$ be a bounded open domain and let $\mathcal{A}(D;\mathbb{R}^{d_a})$ and $\mathcal{U}(D;\mathbb{R}^{d_u})$ be Banach spaces of functions (KREYSZIG, 1978). The target operator is a (potentially nonlinear) map

$$G^\dagger : \mathcal{A} \to \mathcal{U}, \qquad a \mapsto u = G^\dagger(a), \tag{2.82}$$

where both the input a and the output u are functions defined on D.

In this work, $G^\dagger$ is the solution operator of the Boussinesq system: given a parameter encoding (c, u_0) containing the PDE coefficients (α, β) and the initial condition u_0, the operator returns the full space-time solution (η, u).

A neural operator G_θ approximates $G^\dagger$ by composing linear integral operators with pointwise nonlinearities. Each linear layer applies an operator of the form:

$$(\mathcal{K}v)(x) = \int_D \kappa(x,y)\, v(y)\, dy, \tag{2.83}$$

where $\kappa : D \times D \to \mathbb{R}^{d_u \times d_u}$ is a learned kernel. The full neural operator architecture stacks these integral layers into the composition:

$$G_\theta = Q \circ \sigma\Big(W_L + \mathcal{K}_L\Big) \circ \cdots \circ \sigma\Big(W_1 + \mathcal{K}_1\Big) \circ P, \tag{2.84}$$

where:

- $P : \mathbb{R}^{d_a} \to \mathbb{R}^{d_c}$ is the lifting operator, mapping the input $a(x)$ pointwise to a higher-dimensional channel representation with $d_c \gg d_a$;
- $\sigma\Big(W_\ell + \mathcal{K}_\ell\Big)$, for $\ell = 1, \ldots, L$, are the operator layers, each combining a local linear map W_ℓ with a nonlocal integral operator $\mathcal{K}_\ell$, followed by a pointwise activation σ;
- $Q : \mathbb{R}^{d_c} \to \mathbb{R}^{d_u}$ is the projection operator mapping the final channel representation back to the output dimension.

Different choices of kernel parametrization for each $\mathcal{K}_\ell$ give rise to different neural operator architectures.

2.4.2 Fourier Neural Operators

Presented by Li et al. (2020), the Fourier Neural Operator (FNO) is the neural operator (2.84) in which each $\mathcal{K}_\ell$ uses a shift-invariant kernel: $\kappa_\ell(x, y) = \kappa_\ell(x - y)$. This assumption turns each integral operator into a convolution, which by the convolution theorem can be evaluated efficiently in the frequency domain.

Each spectral layer updates the channel representation via:

$$v_{\ell+1}(x) = \sigma\Big((W_\ell\, v_\ell)(x) + (\mathcal{K}_\ell v_\ell)(x)\Big). \tag{2.85}$$

The operator $\mathcal{K}_\ell$ acts via convolution with the shift-invariant kernel κ_ℓ:

$$(\mathcal{K}_\ell v_\ell)(x) = \int_D \kappa_\ell(x - y)\, v_\ell(y)\, dy. \tag{2.86}$$

By the convolution theorem, this equals:

$$(\mathcal{K}_\ell v_\ell)(x) = \mathcal{F}^{-1}\Big(\mathcal{F}(\kappa_\ell) \cdot \mathcal{F}(v_\ell)\Big)(x). \tag{2.87}$$

To see how this is parametrized, recall that for a periodic function f on $[0, 2\pi]$:

$$f(x) = \sum_{k\in\mathbb{Z}} \hat{f}(k)\, e^{ikx}, \qquad \hat{f}(k) = \frac{1}{2\pi}\int_0^{2\pi} f(x)e^{-ikx}\, dx. \tag{2.88}$$

The convolution $(\kappa * v)$ simplifies to:

$$(\kappa * v)(x) = \int_0^{2\pi} \kappa(x - y)\, v(y)\, dy = \sum_{k\in\mathbb{Z}} \big(2\pi\, \hat{\kappa}(k)\, \hat{v}(k)\big)\, e^{ikx}. \tag{2.89}$$

Instead of learning κ_ℓ as a function in physical space, the FNO directly parametrizes its Fourier coefficients as learnable complex-valued matrices $R_{\ell,k} \approx 2\pi\, \hat{\kappa}_\ell(k)$. Only the

$|k| \leq k_{\max}$ low-frequency modes are retained, with all others set to zero; $k_{\max}$ is a hyperparameter.

Training Objective

The FNO is trained as a pure supervised learning problem. Given the dataset $\mathcal{D}$ (2.27) of parameter-solution pairs, the training objective minimizes the mean squared error between the predicted and reference space-time fields:

$$\mathcal{L}_{\mathrm{FNO}}(\theta) = \frac{1}{M} \sum_{i=1}^{M} \frac{1}{N_x N_t} \sum_{j=0}^{N_x - 1} \sum_{n=0}^{N_t} \Big[(\eta_\theta^i(x_j, t_n) - \eta_i(x_j, t_n))^2 + (u_\theta^i(x_j, t_n) - u_i(x_j, t_n))^2 \Big], \tag{2.90}$$

where $(\eta_\theta^i, u_\theta^i) \coloneqq G_\theta(\alpha_i, \beta_i)$ is the network prediction for sample i.

Application to Non-Time-Periodic Problems

When the solution is also assumed to be periodic in time with period T (so that $D = [0, 2\pi] \times [0, T]$), shift-invariance extends to the time direction as well: $\kappa_\ell(x, y, t, \tau) = \kappa_\ell(x - y,\, t - \tau)$. The convolution then generalizes to:

$$(\mathcal{K}_\ell v_\ell)(x, t) = \int_D \kappa_\ell(x - y,\, t - \tau)\, v_\ell(y, \tau)\, dy\, d\tau = \mathcal{F}^{-1}\Big(\mathcal{F}(\kappa_\ell) \cdot \mathcal{F}(v_\ell)\Big)(x, t), \tag{2.91}$$

and the FNO parametrizes the two-dimensional Fourier modes (k_x, k_t) with $|k_x| \leq k_{x,\max}$ and $|k_t| \leq k_{t,\max}$, both hyperparameters.

For problems that are not time-periodic, however, applying the Fourier transform along the temporal axis implicitly extends the signal periodically, potentially introducing a jump discontinuity at the temporal boundary. This can trigger the Gibbs phenomenon (GIBBS, 1899), producing oscillatory artifacts near the boundary that persist regardless of how many temporal modes are retained. Standard signal-processing techniques such as zero-padding and window functions (e.g. Hann, Hamming windows) can partially mitigate these effects, at the cost of altering the temporal structure of the signal. Li et al. (2020) acknowledge this limitation and provide an alternative formulation in which the Fourier layers operate only along the spatial axis while the model advances forward in time, avoiding the periodicity assumption in the temporal direction entirely. Since the Boussinesq problems studied here do not exhibit time periodicity, we adopt this strategy, treating time as a marching direction rather than a spectral domain.

2.4.3 Physics-Informed Neural Operators

Analogously to how PINNs add a physics-informed residual loss to the MLP training objective, the Physics-Informed Neural Operator (PINO) (LI et al., 2023a) incorporates PDE constraints into the FNO training. The FNO backbone remains unchanged; what differs is the loss function, which now combines a data term with residuals evaluated on the predicted space-time fields.

Unlike a PINN, which represents the solution as a single continuous function and differentiates it via automatic differentiation, the FNO outputs (η_θ, u_θ) as a discrete $N_x \times N_t$ array in a single forward pass over the full space-time domain. Evaluating the PDE residuals therefore reduces to numerically differentiating this output array.

Spatial derivatives are computed pseudospectrally using ξ_k (2.18): the DFT of η_θ along the spatial axis at each time t_n yields $\hat{\eta}_\theta(k, t_n)$, from which

$$(\eta_x)_j = \mathcal{F}^{-1}\big(i\,\xi_k\,\hat{\eta}_\theta(k,t_n)\big)_j\,, \qquad (\eta_{xx})_j = \mathcal{F}^{-1}\Big(-\xi_k^2\,\hat{\eta}_\theta(k,t_n)\Big)_j\,, \tag{2.92}$$

and identically for u_θ. Time derivatives are computed by finite differences: second-order central in the interior and first-order one-sided at the boundaries,

$$(\eta_\theta)_t^n = \begin{cases} \dfrac{\eta_\theta^{n+1} - \eta_\theta^n}{\Delta t}, & n = 0, \\ \dfrac{\eta_\theta^{n+1} - \eta_\theta^{n-1}}{2\,\Delta t}, & 1 \le n \le N_t - 1, \\ \dfrac{\eta_\theta^{n} - \eta_\theta^{n-1}}{\Delta t}, & n = N_t, \end{cases} \tag{2.93}$$

and identically for $(u_\theta)_t^n$. The mixed term $(u_\theta)_{xxt}$ is obtained by first computing $(u_\theta)_{xx}$ spectrally and then applying the time-difference stencil to the result. This combination of spectral spatial derivatives and finite-difference temporal derivatives follows the reference implementation of Li et al. (2023a) (LI et al., 2023b).

With all derivatives available, the Boussinesq equations (2.1) define two differential operators acting on the predicted fields:

$$\mathcal{D}_1(\eta_\theta, u_\theta) \coloneqq (\eta_\theta)_t + (u_\theta)_x + \alpha\,(\eta_\theta\, u_\theta)_x, \tag{2.94}$$

$$\mathcal{D}_2(\eta_\theta, u_\theta) \coloneqq (u_\theta)_t - \tfrac{\beta}{3}(u_\theta)_{xxt} + (\eta_\theta)_x + \alpha\,(u_\theta (u_\theta)_x). \tag{2.95}$$

Here $(\eta_\theta^i, u_\theta^i) \coloneqq G_\theta(\alpha_i, \beta_i)$ denotes the network output for sample i, and the subscript $_{j,n}$ indicates evaluation at the grid point (x_j, t_n). The PINO loss is a weighted sum of three terms:

$$\mathcal{L}_{\text{PINO}}(\theta) = \lambda_{\text{pde}}\,\mathcal{L}_{\text{pde}}(\theta) + \lambda_{\text{ic}}\,\mathcal{L}_{\text{ic}}(\theta) + \lambda_{\text{data}}\,\mathcal{L}_{\text{data}}(\theta), \tag{2.96}$$

where:

- $\mathcal{L}_{\text{pde}}(\theta) = \dfrac{1}{M}\displaystyle\sum_{i=1}^{M}\dfrac{1}{N_xN_t}\sum_{j=0}^{N_x-1}\sum_{n=0}^{N_t}\Big[\mathcal{D}_1(\eta_\theta^i, u_\theta^i)_{j,n}^2 + \mathcal{D}_2(\eta_\theta^i, u_\theta^i)_{j,n}^2\Big]$ penalizes violation of the Boussinesq equations at each grid point, with residuals evaluated using the parameters (α_i, β_i) of each sample;

- $\mathcal{L}_{\text{ic}}(\theta) = \dfrac{1}{M}\displaystyle\sum_{i=1}^{M}\dfrac{1}{N_x}\sum_{j=0}^{N_x-1}\Big[(\eta_\theta(x_j,0) - \eta_0(x_j))^2 + (u_\theta(x_j,0) - u_0(x_j))^2\Big]$ enforces the shared initial condition (2.1);

- $\mathcal{L}_{\text{data}}(\theta) = \dfrac{1}{M}\displaystyle\sum_{i=1}^{M}\dfrac{1}{N_xN_t}\sum_{j=0}^{N_x-1}\sum_{n=0}^{N_t}\Big[(\eta_\theta^i(x_j,t_n) - \eta_i(x_j,t_n))^2 + (u_\theta^i(x_j,t_n) - u_i(x_j,t_n))^2\Big]$ measures deviation from the reference solutions $(\eta_i, u_i) \in \mathcal{D}$ (2.27);

- $\lambda_{\text{pde}}, \lambda_{\text{ic}}, \lambda_{\text{data}} > 0$ are weighting hyperparameters.

The PINO therefore generalizes the FNO by adding physics supervision: it can be trained with fewer labeled samples by relying on the PDE residual for additional guidance.

3 METHODOLOGY

4 NUMERICAL RESULTS

quick notes:
fno tem gibbs
gibbs some em PINO
PINO sem dados tem vies espectral

5 CONCLUSION

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